

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

May 2013

CL 103.01 Mechanics of Solids

Date: 03.05.2013, Friday

Time: 1:30p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Figures shown are not to scale.

SECTION - I

Q-1 State and explain: (i) Lami's theorem (ii) Principle of transmissibility [05]

(a)

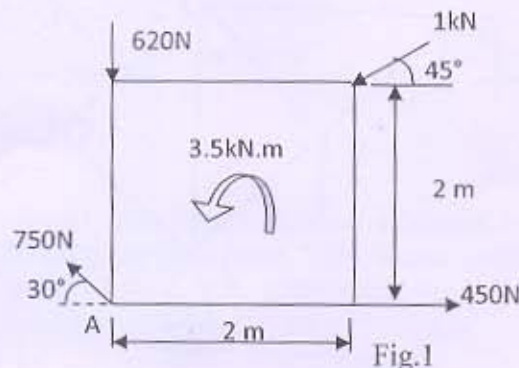
OR

(a) Explain in brief: Fixed vector, Free vector, Force, Rigid body & deformable body [05]

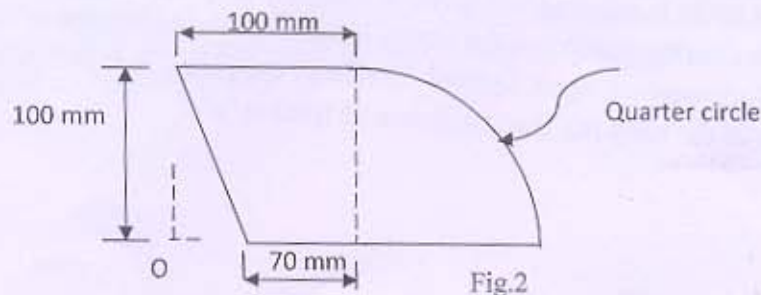
(b) 180N pull force and 350N push force is acting on a body. If the angle between the pull and the push is 135° , then find the magnitude and direction of the resultant force. [02]

Q - 2 Find the Resultant for the force system shown in Fig.1. Locate the same with respect to point 'A'. [07]

(a)



(b) Determine the centroid with respect to point 'O' for the plane area shown in Fig.2. [07]



OR

Q - 2 Reduce the concurrent force system shown in Fig.3 into a single force.

[07]

(a)

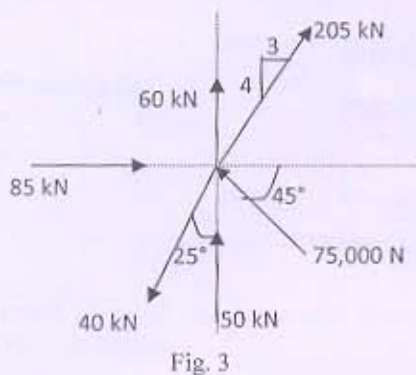


Fig. 3

(b) Derive the coordinates of centroid for quarter circular arc having radius 'R'.

[07]

Q - 3 A weightless ladder, of length 8m, is resting against a smooth vertical wall and rough horizontal ground making 30° with the wall. The coefficient of friction between the ground and the ladder is 0.25. A man weighing 500N wants to climb up the ladder. Find

[07]

(a) how much distance along the ladder the man can climb without slip. A second person weighing 800N wants to climb up the same ladder. Would he climb less than the earlier person? Find his distance covered.

(b) Find the forces in members CE, AE & AD of the truss as shown in Fig. 4.

[07]

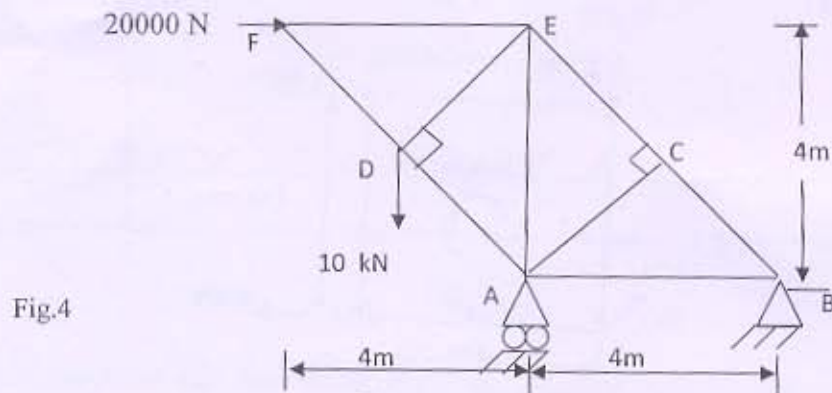


Fig.4

OR

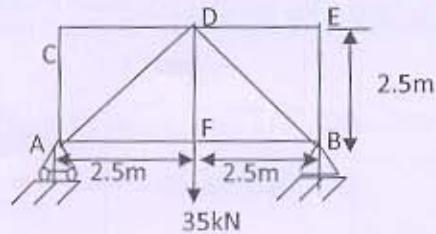
Q - 3 An effort of 200N is required just to move a certain body up an inclined plane of an angle 15° , the force acting parallel to the plane. If the angle of inclination of the plane is made 20° , the effort required, again applied parallel to the plane, is found to be 250N. Find (i)

[07]

(a) The weight of the body (ii) The coefficient of friction ' μ '.

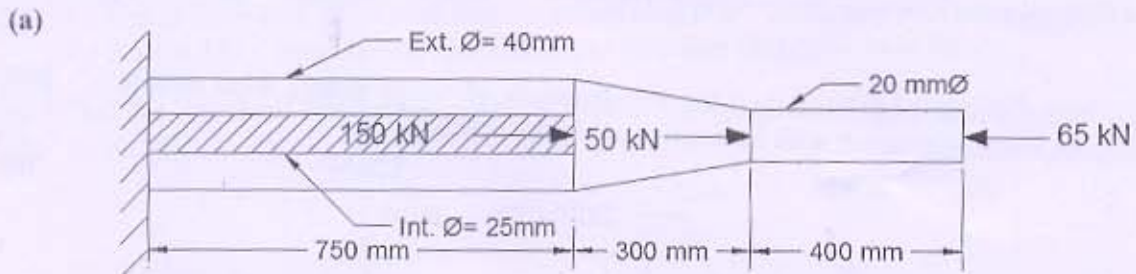
- (b) Identify 'zero force' members of the truss and also determine forces in all other members of the truss as shown in Fig. 5. [07]

Fig.5



SECTION - II

- Q - 4 Derive formula from first fundamental to determine extension ΔL of uniformly tapering circular bar of diameter d_1 at one end, d_2 at other end. The bar has length L , Modulus of Elasticity E and is subjected to axial tensile force P . [05]
- (a)
- (b) Define: 1) Ductility 2) Poisson's ratio 3) Factor of Safety 4) Working Stress 5) Malleability [05]
- Q - 5 Determine net deformation of the bar shown in figure. Take $E = 105 \text{ GPa}$ [06]

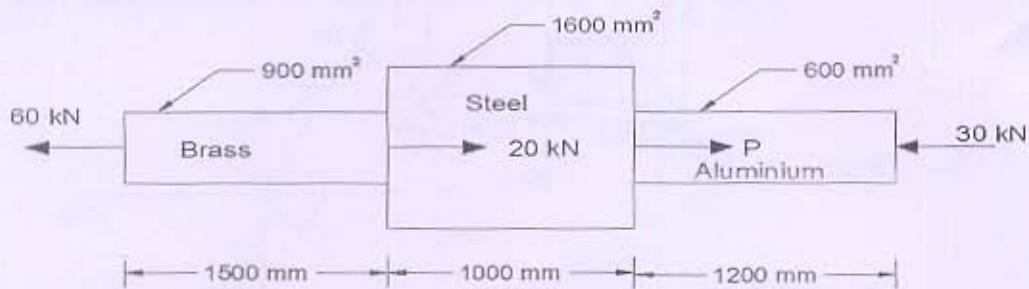


- (b) A concrete column of cross section $300\text{mm} \times 300\text{mm}$ is reinforced by four no. of steel bars of 20mm diameter at its corners. The column carries an axial compression of 400 kN . Check the safety of the section if allowable stress in concrete is 5 N/mm^2 and in steel is 250 N/mm^2 . Take $E_s = 200 \text{ GPa}$, $E_c = 2.5 \times 10^4 \text{ N/mm}^2$ [05]

OR

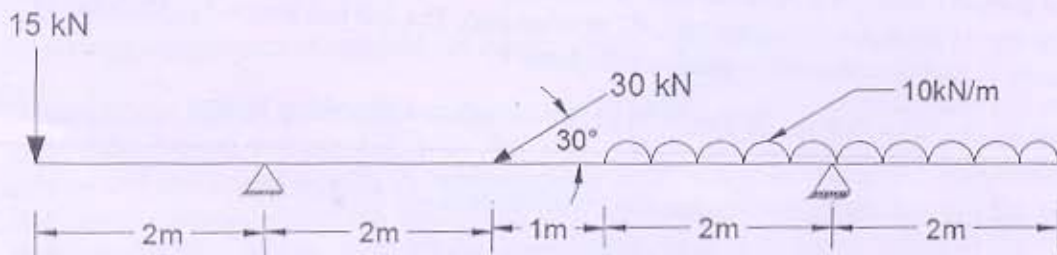
- Q - 5 A flat bar of copper is sandwiched between two flat aluminium bars. The bar dimensions are $1000\text{mmL} \times 100\text{mmW} \times 10\text{mm}$ thick and $1000\text{mmL} \times 100\text{mmW} \times 12\text{mm}$ thick for aluminium bars and copper bar respectively. Calculate stresses in each bar as the temperature of composite bar is increased by 70°C . Take $E_c = 120 \text{ GPa}$, $E_{al} = 80 \text{ GPa}$, $\alpha_c = 17 \times 10^{-6} / ^\circ\text{C}$ and $\alpha_{al} = 24 \times 10^{-6} / ^\circ\text{C}$ [05]
- (a)

- (b) Calculate the force 'P' required for equilibrium of bar shown in Figure. Determine total change in length of the bar. $E_s = 200 \text{ GPa}$, $E_b = 100 \text{ GPa}$, $E_{al} = 75 \text{ GPa}$ [06]

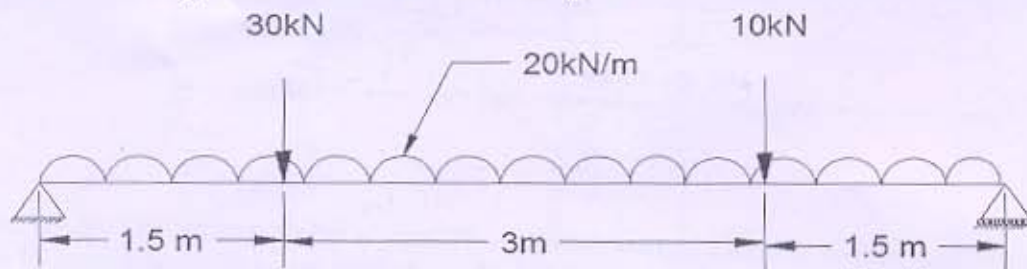


- Q - 6 Determine support reactions and draw Shear Force and Bending Moment diagrams of beam shown below. [08]

(a)



- (b) Draw Shear Force and Bending Moment diagrams of given beam. Also determine maximum bending moment with its location along the beam. [06]



OR

- (b) Determine support reactions and draw Shear Force and Bending Moment diagrams of fixed beam shown below. [06]

